Vipul Sapra

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EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Business Intelligence Engineering, Predictive Modeling, Intent Scoring & Conversion Prediction, Dimensional Data Modeling, Star/Snowflake Schemas, ETL Pipeline Design, Automated Data Quality Audits, Statistical Analysis, Feature Engineering, Production Deployment, Agile Project Delivery

Programming Languages & Tools: SQL, Python, PySpark, Apache Airflow, Tableau, Power BI, AWS (S3, Redshift), Azure (Data Factory, Synapse, SQL), Databricks, Scikit-Learn, XGBoost, Excel (Advanced)

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Senior Business Intelligence Engineer & Applied ML Specialist — **May 2022 – Present**

- **Architected automated intelligence pipelines and semantic layers** utilizing advanced SQL, Python, and Airflow to ingest large-scale engagement and operational signals, driving automated reporting and reducing manual turnaround by over 80%.
- **Developed and deployed production scoring and prioritization models** using XGBoost, Random Forests, and Scikit-Learn for intent scoring and customer behavior prediction, improving system reliability by 20% and empowering account and operations teams with actionable insights.
- **Designed executive dashboards and knowledge repositories** leveraging Tableau, Power BI, and automated QA testing frameworks (SpectrumIQ) to convert complex datasets into digestible business metrics, shaping key operating metrics (KOM) for leadership decision-making.
- **Engineered high-throughput dimensional data models and ETL workflows** incorporating Star/Snowflake schemas and Slowly Changing Dimensions (SCD Type 1/2) across Azure and AWS data warehouses to support large-scale business operations and maintain 99.9% data availability SLAs.
- **Implemented AI-assisted development tools and conversational frameworks** using prompt engineering and automated regex classification to accelerate delivery cycles by 40% while streamlining end-to-end project execution and reporting workflows.

Graduate Research & Information Systems Analyst — **May 2022 – Present**

- **Built advanced statistical and predictive models** utilizing Python data manipulation packages (Scikit-Learn, Pandas) to analyze complex datasets and uncover actionable business opportunities.
- **Designed scalable relational databases and robust SQL queries** to extract, transform, and evaluate data supporting academic and applied information systems research.
- **Synthesized and presented complex analytical findings** to stakeholders, translating technical data insights into strategic business recommendations.

Petrochemical Engineering Analyst — **May 2022 – Present**

- **Analyzed complex operational datasets** to identify process bottlenecks and drive efficiency improvements across industrial workflows.
- **Developed data collection and monitoring frameworks** to track key performance indicators and ensure adherence to quality and reliability standards.
- **Collaborated with cross-functional technical teams** to optimize resource allocation and execute end-to-end operational projects.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.